



**MANIPAL
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AI Teaching Assistant Using the Socratic Method for Conceptual Learning

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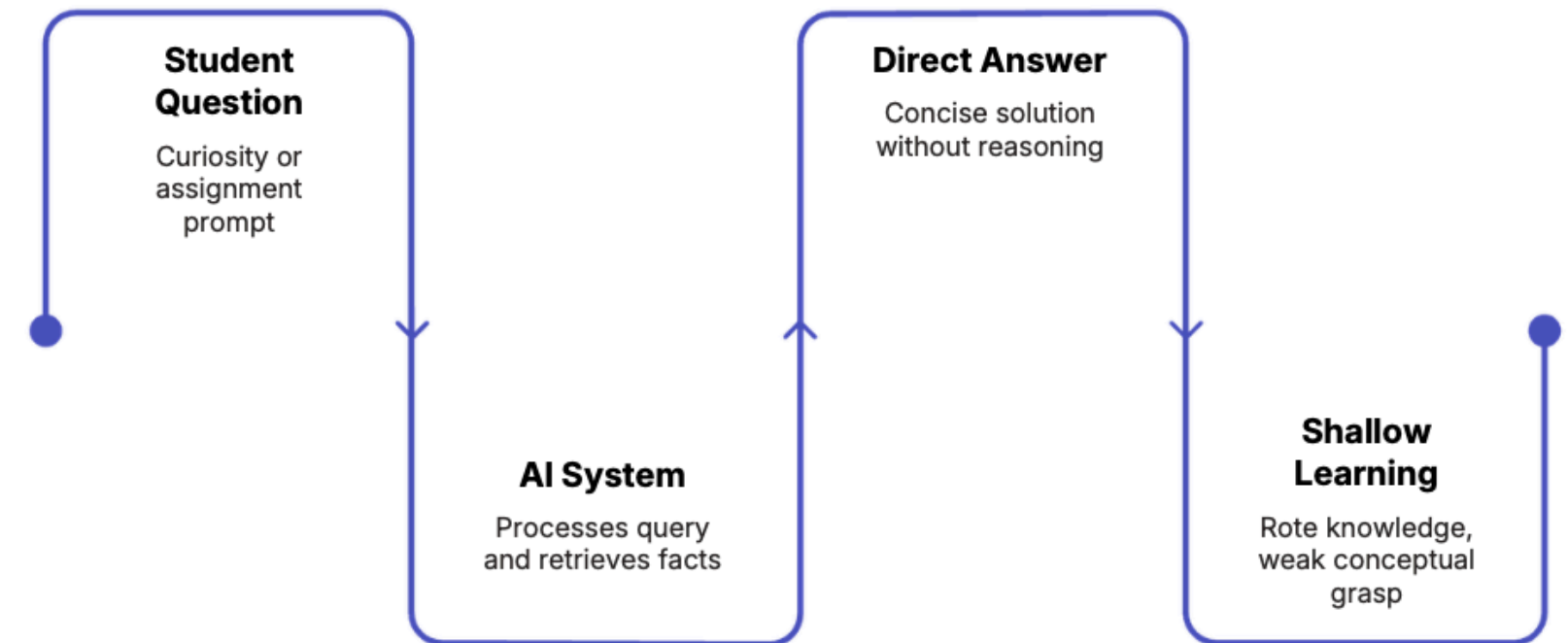
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Dr. Amit Kumar Gupta**

Introduction

Problem Context

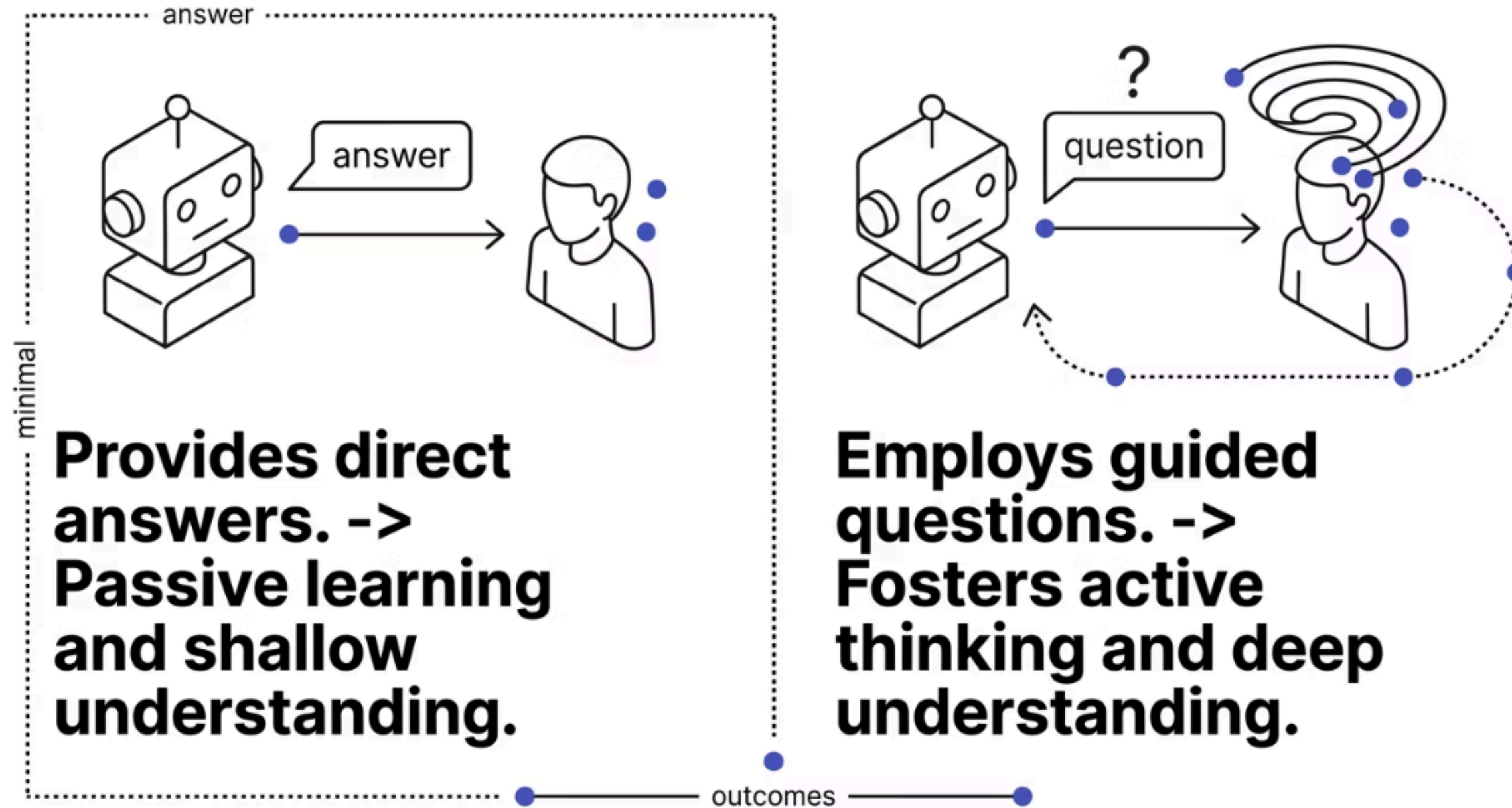
- Rise of AI in education
- Direct answers lead to rote learning
- Weak conceptual clarity
- Lack of critical thinking

Current Learning Flow



This linear flow often leads to superficial understanding.

Learning Through Questioning



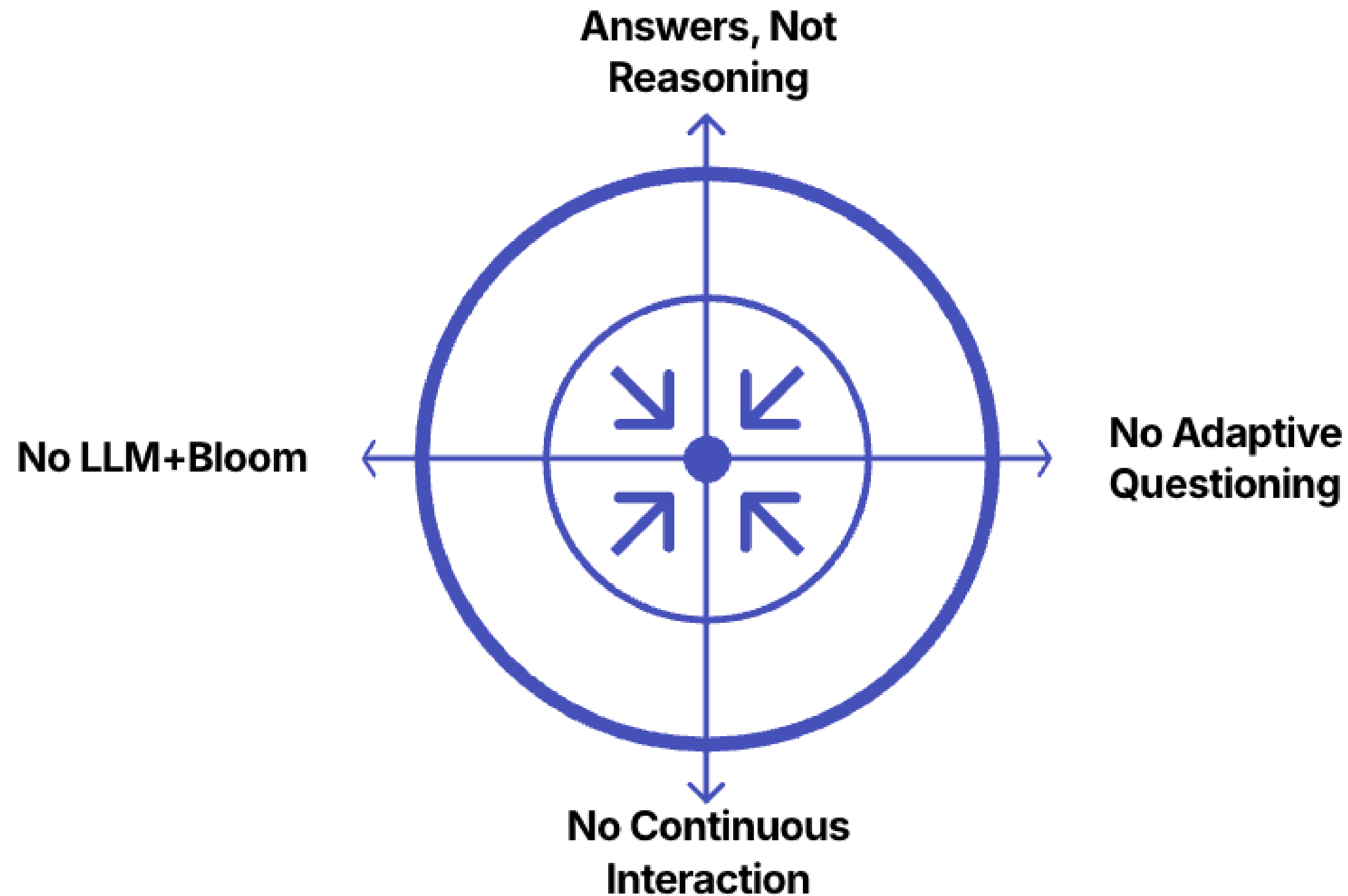
Key benefits listed:

- Encourages reasoning
- Improves conceptual clarity
- Develops critical thinking

Related Work & Research Papers

Paper	Approach	Limitation
Automatic Question Generation and Answer Assessment: A Survey (2021)	Survey (Rule + Neural)	No adaptive difficulty
A Survey of Approaches to Automatic Question Generation (2021)	Rule/Template/Neural	No difficulty control
Automatic Question Generation: A Review (2023)	Transformer models	Weak evaluation metrics
Towards Automatic Question Generation Using Pre-trained Language Models (2024)	T5/GPT	Limited generalization
Automatic Question-Answer Pair Generation in Education (2024)	QA pipeline	Weak conceptual questions
A Survey on Neural Question Generation (IJCAI 2024)	Neural/Transformer	No dialogue integration
Automating Question Generation from Educational Text (2023)	Curriculum-aware	No real-time interaction
Automatic Question Generation using Machine Learning (2022)	ML/Deep Learning	Language limitations
Controllable Open-ended Question Generation (2021)	Controlled generation	No learner feedback
Future of Learning in the Age of Generative AI (2024)	Conceptual	No implementation

Identified Gaps in Current Systems



Research Gap: No system currently combines **Socratic questioning** + **adaptive learning** + **conversational AI** for conceptual learning.

Problem Statement

Create a clear problem definition:

Students rely on direct-answer AI systems

Leads to shallow understanding and weak conceptual clarity

Need AI that guides thinking instead of providing answers

Project Objectives



Build Socratic AI tutor



**Use Bloom's Taxonomy for
question progression**



**Evaluate student responses
semantically**

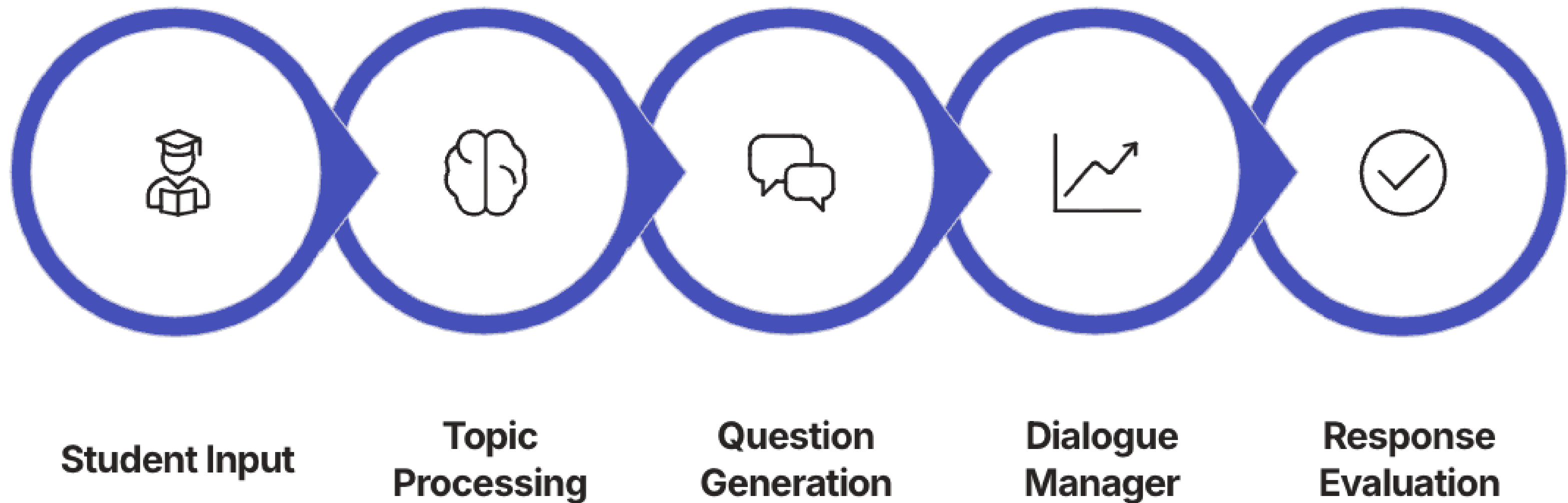


**Adapt difficulty dynamically based on
performance**



**Create conversational system with dialogue
management**

System Architecture



This diagram illustrates the complete pipeline of the AI teaching assistant, starting with **Student Input** which undergoes **Topic Processing**. Questions are then formulated by the **Question Generation (LLM)** module, managed by the **Dialogue Manager**, and assessed by **Response Evaluation**. The **Adaptive Controller** adjusts the system based on these evaluations, looping back to further **Student** interaction.

Socratic Question Types

Here are five types of Socratic questions designed to stimulate critical thinking and deeper understanding:



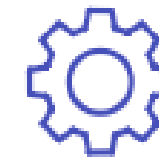
Clarification Questions

"What do you mean by...?"



Why Questions

"Why do you think that?"



Application Questions

"How would you apply this?"



Counterexamples

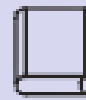
"What if...?"



Reflection Questions

"What did you learn?"

Technical Approach



Bloom's Taxonomy Progression

For question level sequencing



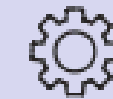
LLM-based Question Generation

Using T5/GPT models



Semantic Similarity Evaluation

For response assessment



Adaptive Difficulty Algorithm

For dynamic adjustment

Progress So Far



Literature Review Completed



Architecture Designed



Question Framework Created



Adaptive Logic Designed



Prototype Planned (in progress)

Outcomes & Next Steps

Achieved

- ✓ Identified research gap
- ✓ Designed Socratic system architecture
- ✓ Created adaptive question framework

Future Work

- → Build and test prototype
- → Integrate LLM (T5/GPT)
- → Conduct user studies with students
- → Evaluate learning outcomes

References

1. Automatic Question Generation and Answer Assessment: A Survey (2021)
2. A Survey of Approaches to Automatic Question Generation (2021)
3. Automatic Question Generation: A Review (2023)
4. Towards Automatic Question Generation Using Pre-trained Language Models (2024)
5. Automatic Question-Answer Pair Generation in Education (2024)
6. A Survey on Neural Question Generation (IJCAI 2024)
7. Automating Question Generation from Educational Text (2023)
8. Automatic Question Generation using Machine Learning (2022)
9. Controllable Open-ended Question Generation (2021)
10. Future of Learning in the Age of Generative AI (2024)

Thank You